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Sard's Theorem

1.1 Set of Measure Zero

Lemma 1.1.1

Suppose $A \subseteq \mathbb{R}^n$ is a compact subset whose intersection with $\{c\} \times \mathbb{R}^{n-1}$ has $(n-1)$ -dimensional measure zero for every $c \in \mathbb{R}$. Then A has a n -dimensional measure zero.

Proof sketch.

把 A 的每一个“片” $A_c := \{(c, x) \in \{c\} \times \mathbb{R}^{n-1} : (c, x) \in A\}$ 的开覆盖, 缩短成里面的一个开的“方片”. 这些“方片”构成 A 的一个开覆盖, 取出一个有限子覆盖计算其 Volume, 将“方片”的相交部分抹去, 使得总体积可控.

Proof. Take $[a, b] \subseteq \mathbb{R}$ such that $A \subseteq [a, b] \times \mathbb{R}^{n-1}$.

Let $A_c := \{(c, x) \in \{c\} \times \mathbb{R}^{n-1} : (c, x) \in A\}$. The projection from A to $[a, b] \times \mathbb{R}^{n-1}$ is continuous map from a compact set to a Hausdorff space, thus is a proper map. It follows that A_c is a compact subspace of A for all $c \in [a, b]$.

Take any $\delta > 0$. There exists a finite open cover $C_1, \dots, C_k$ of A_c , such that the total volume is less than δ . Denote the open subset $C_1 \cup \dots \cup C_k \subseteq A_c$ by U_c . We claim that there is an open interval J_c containing c and contained in U_c , such that A_c is contained in the intersection of A with $J_c \times \mathbb{R}^{n-1}$: otherwise, there is a sequence $\{(c_i, x_i)\}$, such that $c_i \rightarrow c$, $x_i \notin U_c$; but by passing to a subsequence $x_i \rightarrow A_c \setminus U_c$, which is a contradiction to $A_c \subseteq U_c$.

The intervals $\{J_c : c \in [a, b]\}$ form an open cover of $[a, b]$, so there is a finite many $c_1 < c_2 < \dots < c_m$ such that $J_{c_1}, \dots, J_{c_m}$ form a finite open cover of A . We may assume that the total length is less than $2|b-a|$ by shrinking the intersection of the members. Then $J_{c_1}, \dots, J_{c_m}$ is an open cover of A with total volume less than $2\delta|b-a|$, which can be arbitrarily small. Hence A has a n -dimensional measure zero. $\square$

Proposition 1.1.2

Suppose A is an open or closed subset of $\mathbb{R}^{n-1}$ or $\mathbb{H}^{n-1}$, and $f : A \rightarrow \mathbb{R}$ is a continuous function. Then the graph of f has measure zero in $\mathbb{R}^n$.

Proof sketch.

对于 A 是紧集的情况, 令 $P^n = \{\text{Sets are } n\text{-dimensional measure zero}\}$. 上面的引理给出了 $\forall c, P_c^{n-1} \implies P^n$ 的结论. 因此降维操作是无损的, 从而归纳法有效.

Proof. First assume that A is compact. We prove the theorem in this case by induction on n . When $n = 1$, it is trivial because the graph of f is at most a point. To prove the inductive step, we appeal to lemma above. For each $c \in \mathbb{R}$, the intersection of the graph of f with $\{c\} \times \mathbb{R}^{n-1}$ is just the graph of the restriction of f to $\{x \in A : x^1 = c\}$, which is in turn the graph of continuous functions of $(n-2)$ variables. It follows by induction that each such graph has $(n-1)$ -dimensional measure zero, and thus by lemma above, the graph of f itself has n -dimensional measure zero.

If A is noncompact, it is a countable union of compact subsets, so the graph of f is a countable union of sets of measure zero. $\square$

Corollary 1.1.3

Every proper affine subspace of $\mathbb{R}^n$ has measure zero in $\mathbb{R}^n$.

Proposition 1.1.4 ▶ Diffeomorphism-invariant of Measure Zero

Supposer $A \subseteq \mathbb{R}^n$ has measure zero and $F : A \rightarrow \mathbb{R}^n$ is a smooth map. Then $F(A)$ has measure zero.

Definition 1.1.5

If M is a smooth n -manifold with or without boundary, we say that a subset $A \subseteq M$ has **measure zero in M** if for every smooth chart (U, φ) for M , the subset $\varphi(A \cap U) \subseteq \mathbb{R}^n$ has n -dimensional measure zero.

Lemma 1.1.6

Let M be a smooth n -manifold with or without boundary and $A \subseteq M$. Suppose that for some collection $\{(U_\alpha, \varphi_\alpha)\}$ of smooth charts whose

domains cover A , $\varphi_\alpha(A \cap U_\alpha)$ has measure zero in $\mathbb{R}^n$ for each α . Then A has measure zero in M .

Proof. Let (V, ψ) be any smooth chart for M . We need to show that $\psi(A \cap V)$ has measure zero. There exists countably many U_α 's covering $A \cap V$. We write

$$\psi(A \cap V \cap U_\alpha) = (\psi \circ \varphi_\alpha^{-1}) \circ \varphi_\alpha(A \cap V \cap U_\alpha)$$

$\varphi_\alpha(A \cap V \cap U_\alpha)$ is a subset of measure zero set $\varphi_\alpha(A \cap U_\alpha)$, thus has measure zero as well. It follows from Proposition 1.1.4 that $\psi(A \cap V \cap U_\alpha)$ has measure zero. Then $\psi(A \cap V)$ is a countable unions of measure zero sets, thus is measure zero as well. $\square$

Exercise 1.1.7

Let M be a smooth manifold with or without boundary. Show that a countable union of sets of measure zero in M has measure zero.

Proof. Suppose $A_1, A_2, \dots$ are sets of measure zero in M . Let $A = \bigcup_{i=1}^{\infty} A_i$. Take charts (U_{ij}, φ_{ij}) , $j = 1, 2, \dots$ such that $A_i \subseteq \bigcup_{j=1}^{\infty} U_{ij}$. Then $\{U_{ij}\}_{i,j}$ are countably many open sets that containing A , such that for each i, j ,

$$\varphi_{ij}(A \cap U_{ij}) \subseteq \bigcup_{i=1}^{\infty} \varphi_{ij}(A_i \cap U_{ij})$$

is a countable union of measure zero sets, thus itself is measure zero. Then it follows from Lemma 1.1.6 that A is of measure zero. $\square$

Proposition 1.1.8

Supposes M is a smooth manifold with or without boundary and $A \subseteq M$ has measure zero in M . Then $M \setminus A$ is dense in M .

Proof. If $M \setminus A$ is not dense. Then A contains a nonempty open subset of M , which implies that there is a chart (V, ψ) , such that $\psi(A \cap V)$ is a nonempty open subsets of $\mathbb{R}^n$. It is impossible that $\psi(A \cap V)$ has measure zero at the same time that it is open. $\square$